

EXPERIMENT 5

Determination and plotting of Electric potential difference between any two points in free space medium separated by a distance R.

Aim:

To write program for Electric potential difference between two points in free space medium using SCILAB.

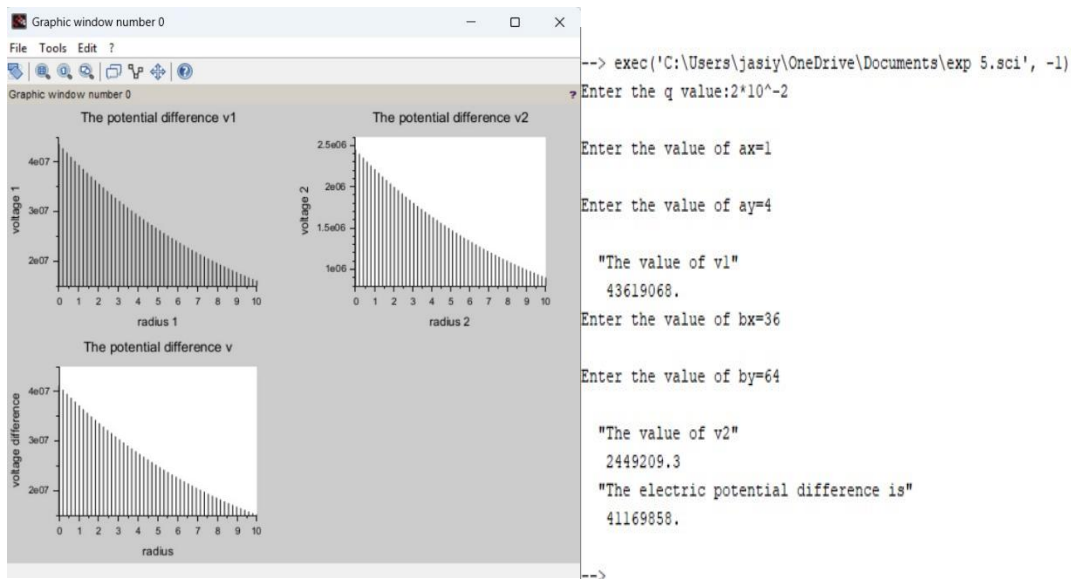
Software required:

- SCILAB version 6.0.1

Code:

```
exp 2 case 1.sc  % exp 3.sc  % exp 3case 1.sc  %
1 clear
2 clc
3 q=input('Enter the q value:');
4 e=8.854*10^-12;
5 pi=3.14;
6
7 ax=input('Enter the value of ax:');
8 ay=input('Enter the value of ay:');
9 x1=sqrt(ax^2+ay^2);
10 v1=q/(4*pi*e*x1);
11 disp('The value of v1',v1);
12
13 bx=input('Enter the value of bx:');
14 by=input('Enter the value of by:');
15 x2=sqrt(bx^2+by^2);
16 v2=q/(4*pi*e*x2);
17 disp('The value of v2',v2);
18
19 x=x1-x2;
20 v=v1-v2;
21 disp('The electric potential difference is',v);
22 x1=linspace(0,10,50);
23 x2=linspace(0,10,50);
24 x=linspace(0,10,50);
25 y=v1*exp(-0.1.*x1);
26 v=v2*exp(-0.1.*x2);
27 x=v*exp(-0.1.*x);
28
29 figure();
30 subplot(2,2,1);
31 plot(x1,y1);
32 xlabel('x radius-1');
33 ylabel('voltage-1');
34 title('The potential difference-v1');
35 subplot(2,2,2);
36 plot(x2,y2);
37 xlabel('x radius-2');
38 ylabel('voltage-2');
39 title('The potential difference-v2');
40 subplot(2,2,3);
41 plot(x,y);
42 xlabel('x radius');
43 ylabel('voltage-difference');
44 title('The potential difference-v');
45
```

Output:



Theoretical Calculation:

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Output:

Enter the q value = 2×10^{-2}

Enter the value ax = 1

Enter the value ay = 4

Enter the value of bx = 36

Enter the value of by = 64

Practical output:

"The value of v1"

43619068.

"The value of v2"

2449209.3

"The electric potential difference is"

41169858.

Theoretical calculation:

Distance of A: $r_1 = \sqrt{1^2 + 4^2} = \sqrt{17} = 4.1231\text{m}$

Distance of B: $r_2 = \sqrt{36^2 + 64^2} = \sqrt{1296 + 4096} = \sqrt{5392} = 73.43$

Electrical potential at A

$V_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{r_1} = \frac{9 \times 10^9 \times 2 \times 10^{-2}}{4.1231} = 4.366 \times 10^7 \text{V}$

$V_1 = 43660000 \text{V}$

Electrical potential at B

$V_2 = \frac{9 \times 10^9 \times 2 \times 10^{-2}}{73.4302} = 2.451 \times 10^6 \text{V}$

$V_2 = 2451000 \text{V}$

Potential Difference

$\Delta V = V_1 - V_2 = 4.366 \times 10^7 - 2.451 \times 10^6$

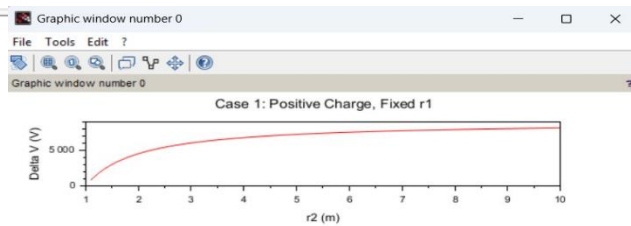
$\Delta V = 4.121 \times 10^7 \text{V}$ or 41210000V

Case 1: A positive charge with fixed $r_1=1$ m r_2 varying.

Code:

```
1 //Constants
2 epsilon0 = 8.854e-12; //Permittivity of free space
3 k = 1./ (4.*%pi * epsilon0); //Coulomb's constant
4 q = 1e-6; //Charge in Coulombs (1 µC)
5
6 //Case 1: Positive charge, r1 = 1 m, r2 varies
7 r1_case1 = 1; //meters
8 r2_case1 = linspace(1.1, 10, 100); //meters
9 deltaV_case1 = k * q * (1./ r1_case1 - 1./ r2_case1);
10
11
12
13 //Plotting
14 clf();
15 subplot(3, 1, 1);
16 plot(r2_case1, deltaV_case1, "r");
17 xlabel("r2 (m)");
18 ylabel("Delta V (V)");
19 title("Case 1: Positive Charge, Fixed r1");
20
21
```

output:

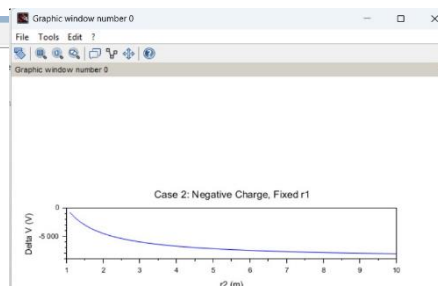


Case 2: A negative charge with r_1 fixed and r_2 varying.

Code:

```
1 // Case 2: Negative charge, r1 Fixed, r2 varies
2 q_case2 = -1e-6; // -1 µC
3 r1_case2 = 1; // meters
4 r2_case2 = linspace(1.1, 10, 100); // meters
5 deltaV_case2 = k * q_case2 * (1./ r1_case2 - 1./ r2_case2);
6 //Plotting
7
8 clf();
9 subplot(3, 1, 2);
10 plot(r2_case2, deltaV_case2, "b");
11 xlabel("r2 (m)");
12 ylabel("Delta V (V)");
13 title("Case 2: Negative Charge, Fixed r1");
14
```

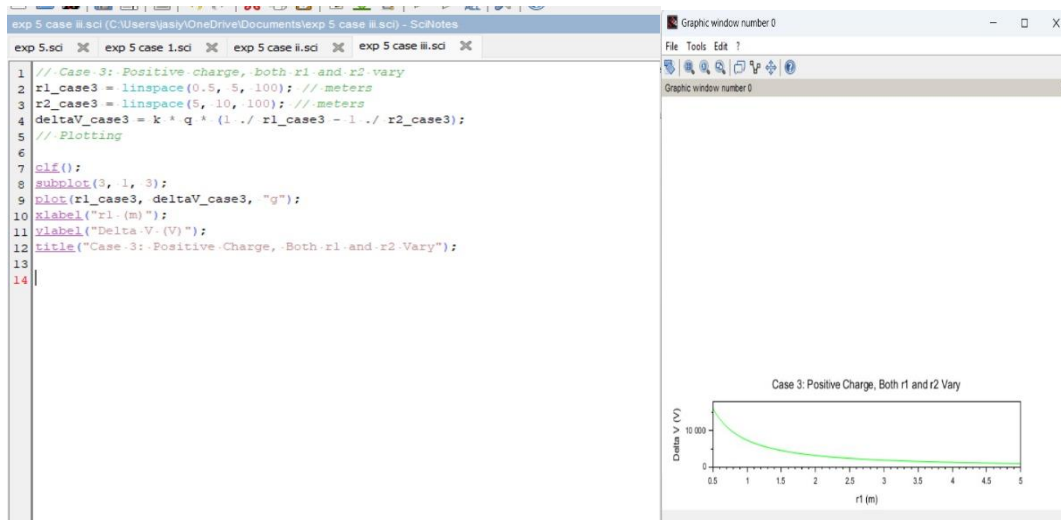
output:



Case 3: Both points r_1 and r_2 vary with a positive charge.

Code:

output:



Result :

Thus the program was verified for Electric potential difference between two points in free space medium using SCILAB was successfully.